

Announcements

- ▶ Catching up on grading: come ask me if you have questions.
- ▶ I will not have my Wed (2–3pm) office hour this week; email me to meet at another time.
- ▶ Schedule:
 - Today (Follow-up)**: Equilibrium temperature without atmosphere
 - Today (New)**: Atmosphere and Greenhouse Effect (week 1)
 - Tomorrow (Lab)**: “feel the sun on your face (no one else can do it for you)”
 - Thursday (Discussion)**: Calculating equilibrium temperatures with atmospheres
- ▶ Next Week: Greenhouse Effect II

Today's Reading Quiz (3 min)

On your own sheet of paper. . .

What are the key physical characteristics of our atmosphere (related to EM radiation, transparency, and opaqueness) that cause the “greenhouse effect”?

Turn in to Blackboard as scanned pdf (or image is fine).

Key information and equations from last week

Wien's Law (wavelength of BB peak)

$$\lambda_{\text{peak}} = \frac{2900 \mu\text{m} \cdot \text{K}}{T}$$

Stefan-Boltzmann Law (total flux output)

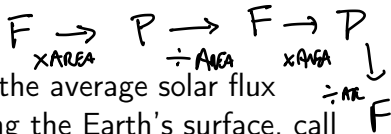
$$F = \left(6 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \cdot \text{K}^4} \right) T^4$$

Peak value of "Spectral Flux" BB curve

$$F_{\lambda} = \frac{F}{1.5 \times \lambda_{\text{peak}}}$$

Object	Characteristic	Size (m)
Sun	Radius, $R_{\odot}$	7.0×10^8
Earth	Distance to Sun, $d_{\odot \rightarrow \oplus}$	1.5×10^{11}
Earth	Radius, $R_{\oplus}$	6.4×10^6

Questions 2–3 — Overview



- ▶ In Question 2, we wish to find the average solar flux (power per area, in $\frac{\text{J/s}}{\text{m}^2}$) reaching the Earth's surface, call it $F_{\text{from Sun}}$.
- ▶ To calculate equilibrium temperature (Question 3), we assume that:

$$F_{\text{from Sun}} = F_{\text{of Earth}}$$

$$F_{\text{from Sun}} = F_{\text{of Earth}}(T_{\text{Earth}})$$

where the flux “from Earth” is its output power per area due to *being a blackbody radiator*.

- ▶ The temperature of the Earth, T_{Earth} , is an unknown at this point (that's what we want to find), but the dependence of the Earth's flux on temperature is just the Stefan-Boltzmann Law:

$$F_{\text{of Earth}}(T_{\text{Earth}}) = \left(6 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \cdot \text{K}^4} \right) T_{\text{Earth}}^4$$

Question 2(a) — Finding Power of the Sun

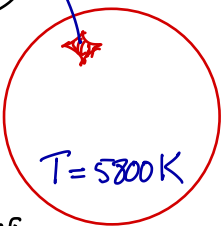
The sun is a blackbody radiator, emitting flux (power per area) according to the Stefan-Boltzmann Law:

$$F = \left(6 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \cdot \text{K}^4} \right) T^4 \quad \xrightarrow{5800\text{K} = T} 6.8 \times 10^7 \frac{\text{J/s}}{\text{m}^2}$$

It's total power output is that flux multiplied by its total surface area (as a sphere, $A = 4\pi R^2$).

$$P_{\text{SUN}} = F_{\text{SUN}} \cdot A_{\text{SUN}}$$

$P = F \cdot A$
 $\frac{\text{J/s}}{(\text{W})} = \frac{\text{J/s}}{\text{m}^2} \cdot \text{m}^2$



$$= \left(6.8 \times 10^7 \frac{\text{J/s}}{\text{m}^2} \right) 4 \cdot \pi \cdot \left(7.0 \times 10^8 \text{m} \right)^2$$

$$= 4.2 \times 10^{26} \frac{\text{J}}{\text{s}}$$

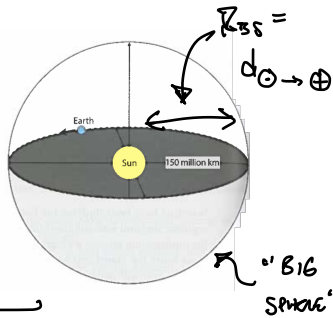
$$= 4.2 \times 10^{26} \text{ W}$$

$$A_{\text{SUN}} = A_{\text{SPHERE}_2} = 4\pi R_{\text{SUN}}^2$$

Question 2(b) — Finding the “Solar Constant”

The so-called “Solar Constant”, S , is the flux (in $\frac{\text{J}}{\text{s m}^2}$) of the Sun at the position of the Earth in orbit.

This is approximately what someone at the Equator would receive if there were no clouds.



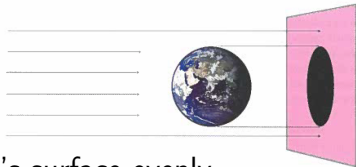
$$S = \left(\text{Flux of Sun Radiation @ Earth Orbit} \right) = \frac{P_{\text{Sun}}}{A_{\text{BIG SPHERE}}}$$

$$= \frac{4.2 \times 10^{26} \text{ W}}{4 \pi R^2} = \frac{4.2 \times 10^{26} \text{ W}}{4 \cdot \pi (1.5 \times 10^{11} \text{ m})^2}$$

$$\approx 1500 \frac{\text{W}}{\text{m}^2} = 1500 \frac{\text{J/s}}{\text{m}^2}$$

Question 2(c) — Finding avg flux at the surface

We obtain an average solar flux (power per area) at the Earth's surface by (i) calculating the total power "caught" by the Earth (see image at right), and (ii) dividing that power over the Earth's surface evenly.



$$P_{\text{SUN CAUGHT BY EARTH}} = "S" \cdot A_{\text{EARTH'S CIRCLE}} = S \cdot \pi R_{\text{EARTH}}^2$$

$$F_{\text{AVG OF SUN ON EARTH}} = \frac{P_{\text{SUN CAUGHT BY EARTH}}}{A_{\text{EARTH}}} = \frac{S \cdot \cancel{\pi R_{\text{EARTH}}^2}}{4 \pi \cancel{R_{\text{EARTH}}^2}} = 1500 \frac{\text{W}}{\text{m}^2} / 4 = 375 \frac{\text{W}}{\text{m}^2} = 375 \frac{\text{J/s}}{\text{m}^2}$$

Question 2(d) — Accounting for Clouds and Ice

A large fraction of the Sun's power is reflected back into space by the Earth. The name of this reflective fraction is the "albedo" and is given by the Greek letter alpha, α . We estimate that $\alpha \approx 0.3 = 30\%$. Therefore the actual average solar flux at the surface, our $F_{\text{from Sun}}$, is ...

$$F_{\text{from Sun @ Earth's Surface (Avg)}} = \frac{S \cdot \pi R_{\text{Earth}}^2}{4 \pi R_{\text{Earth}}^2} \underbrace{(1 - \alpha)}_{\text{Fraction Absorbed (Not Reflected)}}$$
$$\approx 260 \frac{\text{W}}{\text{m}^2} = 260 \frac{\text{J/s}}{\text{m}^2}$$

Question 3 — Estimating T_{Earth}

- (a) Draw an energy flow diagram for the Sun-Earth system.
- (b) If we assume the Earth is in equilibrium (“total” energy is constant, so T is constant), what equation do we obtain? What is the equilibrium value of the Earth’s temperature?
- (c) Is this answer correct? Why or why not?

$$F_{\text{SUN}, \text{IN}} = F_{\text{EARTH}, \text{OUT}}(T_{\text{EARTH}})$$

$$F_{\text{SUN}, \text{IN}} = \left(6 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \cdot \text{K}^4} \right) T_{\text{EARTH}}^4$$

$$T_{\text{EARTH}} = \left[F_{\text{SUN}, \text{IN}} / (\quad) \right]^{1/4} = 257 \text{ K} \rightarrow -16^\circ \text{C}$$

Now, what about with an atmosphere?

The details of our atmosphere are approximately:

- ▶ The atmosphere is *transparent* to the Sun's visible light.
(So our prior calculations are unchanged)
- ▶ The atmosphere is *opaque* to (far) infrared light, like the Earth's thermal radiation at $\sim 15^{\circ}\text{C}$.
- ▶ The atmosphere is a blackbody, radiating up and down.

Let's make an energy flow diagram...